

EECS 16A Designing Information Devices and Systems I

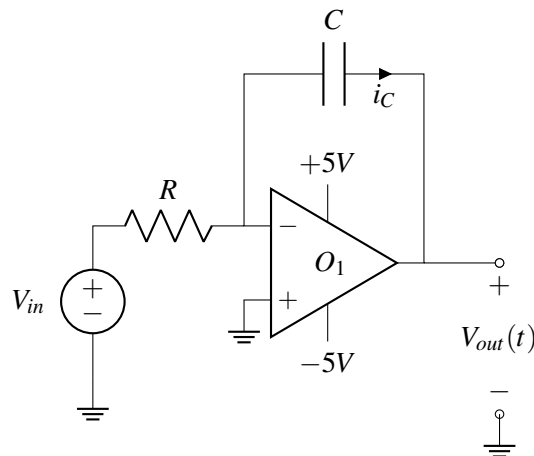
Discussion 12B

1. Integration using Op-amps

As we have seen already, it is useful in several applications to create triangular wave voltages (also known as voltage ramps). For example, in the capacitive touchscreen, we charged a capacitor using a constant current source that flipped from positive current to negative current periodically, which created triangular waveforms for the voltage on the capacitor. Then we compared the capacitor's voltage to a reference voltage to detect touch. In this problem, you will be analyzing a circuit that produces voltage ramps using a voltage source and an op-amp in negative feedback.

- (a) One of the circuit blocks you can use to generate the triangular waveform is the integrator. An integrator outputs the integral of the input signal. For the circuit given below express $V_{out}(t)$ in terms of R , C , V_{in} , and t , assuming V_{in} is not varying with time. What is the slope of $V_{out}(t)$? You may also assume that capacitor C has 0 V across it at time $t = 0$.

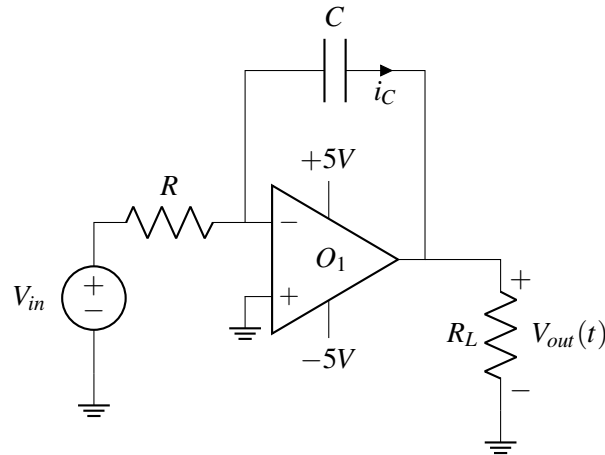
Hint: You will have to apply KCL, and use the fact that the current flowing through a capacitor is given by $I = C \frac{dV}{dt}$.



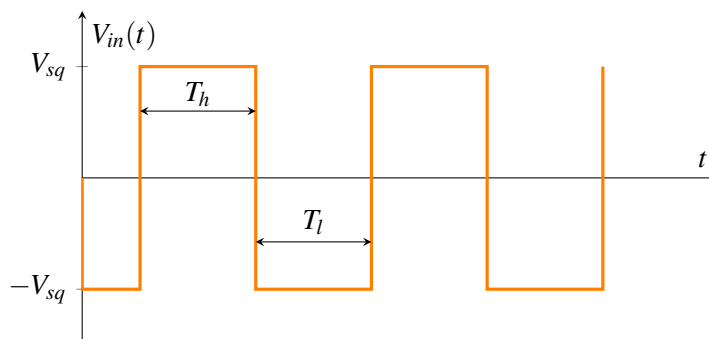
- (b) What is the value of the current i_C flowing through capacitor C , in terms of V_{in} , R and C ? How does the capacitor current change if we double C ? How does the slope of $V_{out}(t)$ change if we double capacitor C ?

Note: the current direction is specified in the figure above.

- (c) If we connect a load resistance at the output of the circuit, as shown in the figure below, does the output voltage $V_{out}(t)$ change, from what you calculated in part (a)? Why or why not?

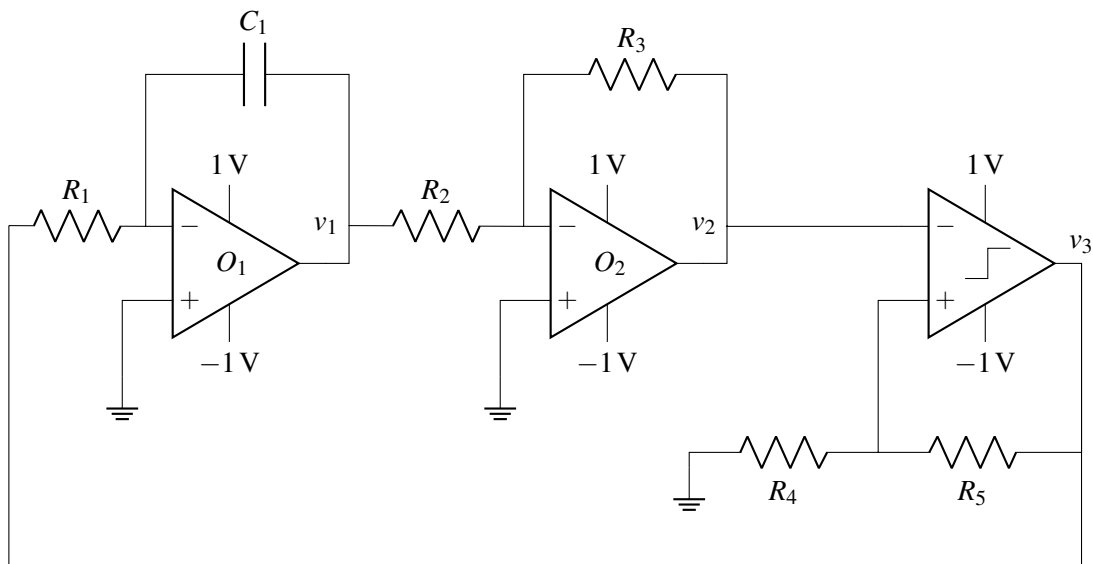


- (d) If $V_{in}(t)$ varies with time as shown in the following diagram, plot $V_{out}(t)$ for $t = 0$ to $t = 1.5T$, where $T = 2T_h = 2T_l$. In your plot indicate an algebraic expression for the slope (as a function of R , C and V_{sq}) and add tick marks on the x and y axis indicating the time and voltage values where the ramp slope changes. You may assume again that capacitor C has 0V across it at time $t = 0$.



2. Timer Circuit

In this problem, we will walk through another useful, real-world circuit, the timer circuit. The circuit is shown below. All resistors have a resistance of $1\text{ k}\Omega$ and $C_1 = 1\text{ }\mu\text{F}$. Assume that all the op-amps are in negative feedback.



- (a) Find the current through the capacitor C_1 in terms of the node voltage v_3 and the resistor R_1 .
- (b) Suppose that at time $t = 0$, C_1 is uncharged. Find the voltage v_1 in terms of t and v_3 . Recall that all resistors have a resistance of $1\text{ k}\Omega$ and $C_1 = 1\text{ }\mu\text{F}$. What is the maximum $|v_1|$ could be?
- (c) How is v_2 related to v_1 ? What is the voltage v_2 ?

Now, let's independently analyze the circuit in the two possible outputs of the comparator, when $v_3 = 1\text{ V}$ and when $v_3 = -1\text{ V}$.

- (d) Assume that the output of the comparator v_3 has railed to the top rail. With this value of v_3 , what is v_2 as a function of time? What is the voltage v_c^+ at the positive input of the comparator? At what time will the two inputs of the comparator be equal?
- (e) Now assume that the reverse occurs, that is, the output of the comparator has railed to the bottom rail. Repeat part (d) with this value of v_3 .
- (f) What is v_3 as a function of time? Draw a graph of v_3 and v_2 . Since the graph is periodic, find its period and frequency.
- (g) Suppose that we changed the value of C_1 to be $2\mu\text{F}$? What is the new period? Suppose that we change R_5 to be $2\text{k}\Omega$. What is the new period? What if we change R_5 to be 0Ω ? Will this circuit still operate?